

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

30th October 2015

Subject ST2 — Life Insurance

Time allowed: Three hours (14.45* – 18.00 Hrs)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. ** You have 15 minutes at the start of the examination in which you are required to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made. You then have three hours to complete the paper.*
3. *You must not start writing your answers in the answer sheet until instructed to do so by the supervisor.*
4. *The answers are not expected to be any country or jurisdiction specific. However, if Examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.*
5. *Attempt all questions, beginning your answer to each question on a separate sheet.*
6. *Mark allocations are shown in brackets.*
7. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

- Q. 1)** A life insurance company has been selling conventional without profit endowment products for the past 15 years. However, the economic outlook has changed significantly in the last two years compared with the time these products were originally launched. It has been decided that the surrender values offered under the product should be reviewed to reflect the changed outlook.
- i)** State the principles which should be taken into account while reviewing the surrender values under a conventional without profits endowment product. (4)
- After the first three years in force, the company calculates surrender values using a prospective formula
- ii)** Explain which elements of the surrender value basis would be most affected by the change in economic outlook. (3)
- [7]
- Q. 2)** **i)** Describe the main features of the following types of life insurance contract:
- Without profits
 - With profits
 - Unit linked
 - Index linked
- (4)
- ii)** Give examples of products typically written under each of the above types. (2)
- iii)** Discuss how the following characteristics of products may differ between the four types in part (i):
- Administration cost
 - Riskiness
 - Profitability
- (6)
- [12]
- Q. 3)** An investor is considering a regular series of investments for 25 years to accumulate into a lump sum which can be used for his retirement. He is considering two with-profit products: under one product the profits are distributed by additions to benefit method; while in the second product the profits are distributed by revalorization method.
- The two products operate in the same market, and are broadly similar in all material respects other than those which would be appropriate to the different methods of profit distribution adopted.
- i)** Describe these two methods of profit distribution, and explain the impact each may have on the potential customer yield. (9)
- ii)** Explain the circumstances under which the revalorization method may provide better return to the investor, assuming that he keeps the policies until maturity. (3)
- [12]

Q. 4) A life insurance company, ABC Life, has been selling individual term insurance business for the last 5 years. ABC Life is now considering entering the employer-employee group life insurance market, where the premiums will be paid by the sponsoring employer and ABC Life will pay a lump sum benefit on death of an employee. This is a large market in the country where the ABC Life operates.

- i) Describe how ABC Life would determine the mortality assumption in the pricing basis. (5)

ABC Life is considering effecting some reinsurance protection on one of the following bases:

- Quota share
- Individual surplus
- Excess of loss

- ii) List reasons why ABC Life might consider reinsurance protection (2)

- iii) Discuss the advantages and disadvantages of each of the reinsurance options that ABC Life is considering. (10)

The following data has been provided for a particular group employer-employee scheme to the company for generating a quotation:

Year	Average No. of members during the year	No. of deaths during the year
201X	200,246	213
201X-1	187,974	169
201X-2	199,783	201
201X-3	156,846	172

- iv) Comment on whether the data appears to be reasonable and identify additional data that would help confirm your conclusion. (6)
[23]

- Q. 5)** i) Define Embedded Value (EV) and its components. (2)

A life insurance company that has been operating for 10 years has never publically published its Embedded Value results so far. The marketing department has suggested that these values be published annually going forward to improve the confidence of investors and policyholders in the company.

- ii) Discuss the factors to be considered while evaluating this suggestion (5)

As part of the published embedded value results, it has also been decided to demonstrate sensitivity of the results. The company sells the following products:

- a) A regular premium pure term insurance product

- b) A unit linked product, with the death benefit, surrender benefit and maturity benefit set equal to the value of the units at the time of death / surrender / maturity
- c) A regular premium pure endowment product with a surrender benefit equal to 70% of the sum of premiums paid.

iii) Outline the expected impact on EV for each of the above products if the assumptions listed below are increased by 50%. The impact of each assumption is to be considered independently. (8)

- a) Mortality
- b) Persistency
- c) Risk Discount Rate

[15]

Q. 6) A life insurance company proposes to launch a new single premium unit-linked endowment life product maturing at age 60 with the following features:

- 99% of the single premium is allocated to units at the offer price.
- Sum assured is payable on death, sum assured is equal to the greater of:
 - the value of units; and
 - x% of the single premium paid, where x% is chosen by the policyholder at outset, with a minimum of 100% and a maximum of 300%.
- No medical underwriting if x% is 100%.
- A management charge of 1% per annum of the value of units for the cost of life cover. The charges and deductions are variable in certain circumstances.
- On attaining age 60 or on earlier surrender, value of units is paid as maturity benefit.

i) Describe the main risks under the product for the insurance company. (10)

ii) Discuss appropriate investments for the assets backing various parts of the liabilities under this policy (5)

It has been suggested to add a guarantee of 100% of the single premium as minimum benefit at maturity to make the product more attractive.

iii) State the main risk added by the introduction of this guarantee (2)

iv) Suggest, with reasons, possible modification to the product design and charge structure which may mitigate this risk. (3)

v) Describe how the cost of providing the guarantee and the suitability of the charging structure for the cost of guarantee can be evaluated using stochastic techniques. (11)

[31]
